C 22571 **(Pages : 2)** **Name.....**

Reg. No.....

FOURTH SEMESTER M.Sc. DEGREE (CBCSS) [REGULAR/ SUPPLEMENTARY] EXAMINATION, APRIL 2022

April 2021 Session for SDE/Private Students

Mathematics

MTH 4E 11—GRAPH THEORY

(2019 Admission onwards)

Time : Three Hours Maximum Weightage : 30

General Instructions

Covid Instructions are not applicable for Pvt/SDE students (April 2021 session)

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section/Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section/sub part/sub division may be ignored.*
4. *There will be an overall ceiling for each Section/Part that is equivalent to the maximum weightage of the Section/Part.*

Part A

Answer **all** *questions.*
Each question carries a weightage 1.

1. Define a Tree.
2. Prove that a set $S \subseteq V$ is an independent set of G if and only if V-S is a covering of G.
3. Prove that a matching M in G is a maximum matching implies that G contains no M-augmenting path.
4. Define a block.
5. Prove that in a critical graph no vertex cut is a clique.
6. Show that if G is K-critical, then $\delta \geq k-1$.
7. State Euler's formula for a connected plane graph.
8. Prove that a connected graph is a tree implies that every edge is a cut edge.

(8 × 1 = 8 weightage)

Turn over

Part B

Answer any **two** *questions from each module.*
Each question carries a weightage 2.

Module I

9. Prove that an edge e of G is a cut edge of G if and only if e is contained in no cycle of G.
10. Prove that a vertex v of G is a cut vertex of G if and only if $d(v) > 1$.
11. If e is a link of G, then prove that $\tau(G) = \tau(G - e) + \tau(G.e)$.

Module II

12. Prove that in a bipartite graph, the number of edges in a maximum matching is equal to the number of vertices in a minimum covering.
13. Show that $\alpha + \beta = \vartheta$.
14. State and prove marriage theorem.

Module III

15. Prove that every k-chromatic graph has atleast k vertices of degree atleast k–1.
16. Prove that K_5 is non-planar.
17. Prove that a graph G is embeddable in the plane if and only if it is embeddable on the sphere.

(6 × 2 = 12 weightage)

Part C

Answer any **two** *questions.*
Each question carries a weightage 5.

18. If G is a simple graph with $\vartheta \geq 3$ and $\delta \geq \frac{\vartheta}{z}$, then prove that G is Hamiltonian.
19. Let G be a bipartite graph with bipartition (X, Y). Then prove that G contains a matching that saturates every vertex in G if and only if $|N(S)| \geq |S|$ for all $S \subseteq X$.
20. State and prove Brooke's theorem.
21. Prove that a non-empty connected graph is Eulerian if and only if it has no vertices of odd degree.

(2 × 5 = 10 weightage)